Data Sufficiency

Vector India Pvt. Ltd.

↳ By Sunil Guptha Yarramalla

↳ The main object of data sufficiency Qs to cross check whether the given data Qs sufficient to answer the given question (or) not.

↳ In D.S [Data Sufficiency] you will be given 2 statements for a question & 5 options.

↳ Mainly we have to cross check the Information in statements Qs sufficient (or) not in such a way that according to the options.

↳ Options are:-

option (A):- Statement I alone sufficient, but not II.

(B):- statement II alone sufficient, but not I.

(C):- Both statements I & II are sufficient, but neither statement alone Qs sufficient.

(D):- Each statement alone Qs sufficient.

(E):- Statements I & II together are not sufficient.

Example Questions:-

Ex 1:- How much Qs 20% of a certain number?

St 1:- 10% of the number Qs 5

St 2:- 40% of twice the number Qs 40.

Ans:- St 1:- 10% of the no. = 5 ⇒ 20% of the no = 10

St 1 Qs sufficient alone

St 2:- 40% of $2x = 40$ ⇒ 20% of $2x = 20$.

20% of $x = \frac{20}{2} = 10$.

(199)

St 2 also alone sufficient.

Option (D) is the answer [each st alone is sufficient].

Ex:2:- Is x greater than 0.27?

St 1:- x is greater than $\frac{1}{4}$

St 2:- x is equals to $\frac{3}{10}$

Ans:- St 1:- $x > 0.25$ => means it may be 0.251 .... 0.27 .... ∞

means for the question we have to give answer as either 100% yes (or) 100% no.

In St:-1:- let us assume x value is 0.251 - 0.27

then answer is x values not greater than 0.27

means answer is no in this assumption.

In St:- 1:- if we assume x value is for values $\geq 0.27 \left[> \frac{1}{4}\right]$ the answer is yes.

But Statements should give only one answer i.e., either yes / not but not both.

In St:-2:- $x = \frac{3}{10} = 0.3$ it is always greater than 0.27 and it is giving as answer yes.

So, Answer is Option (B).

Ex:-3:- What is the value of sum of a list of n odd integers?

St:1:- n=8.

St:-2:- The square of the number integers in the list is 64.

Ans:- First of all we must know those list of odd integers are what kind, means first odd integers (or) first odd integers divisible by certain number.

200

then we must know how many odd integers from the statements.

1) In this question only data is sufficient to find no. of odd integers i.e., 8 through the information from statements. but, we dont have which of those 8 odd integers information.

So, we can't find out solution from both the stmts.

Answer is Option (E)

Ex: 4:- If $5x+3y=17$, what is the value of x?

St 1:- x is a +ve integer

St 2:- $y=4x$.

Ans:- from St 1:- we can get property of x but not its value. so, not sufficient.

from st 2:- $y=4x$, if we sub this in the given equation. $5x+3y=17$.

$5x+3(4x)=17 \Rightarrow 17x=17 \Rightarrow x=1$ = +ve only

So, only st 2 alone sufficient.

Option [B]

Ex: 5:- Is x a negative number?

St 1:- $9x>10x$.

St 2:- $x+3$ is +ve.

Ans:- from st 1:- $9x>10x$ is possible only when 'x' is -ve integer. so, its alone sufficient.